

JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

CLASS

BIOLOGY

9744/01

Paper 1 Multiple Choice

15 September 2017

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name and class on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

This document consists of **18** printed pages and **2** blank pages.

[Turn over

- 1 An actively growing cell is supplied with radioactive amino acids.

Which cell component would first show an increase in radioactivity?

- A Golgi body
- B mitochondrion
- C nucleus
- D rough endoplasmic reticulum**

- 2 When mucus is secreted from a goblet cell in the trachea, these events take place.

- 1 addition of carbohydrate to protein
- 2 fusion of the vesicle with the plasma membrane
- 3 secretion of a glycoprotein
- 4 separation of a vesicle from the Golgi body

What is the sequence in which these events take place?

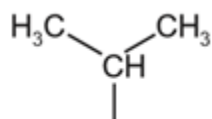
- A 1 → 4 → 2 → 3**
- B 1 → 4 → 3 → 2
- C 4 → 1 → 2 → 3
- D 4 → 1 → 3 → 2

- 3 Which combination is found in a prokaryotic cell?

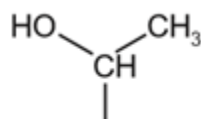
	endoplasmic reticulum	DNA	RNA	nucleus
A	X	✓	X	X
B	✓	X	X	✓
C	X	✓	✓	X
D	X	X	✓	✓

- 4 Threonylvaline is a dipeptide formed from the two amino acids, valine and threonine. A peptide bond forms between the amine group of valine and carboxyl group of threonine.

The side-chains (R groups) of the two amino acids are shown.

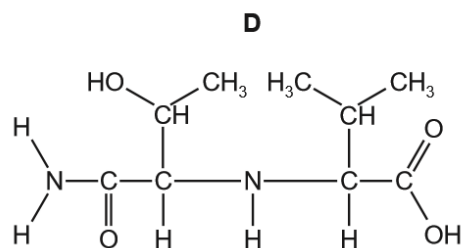
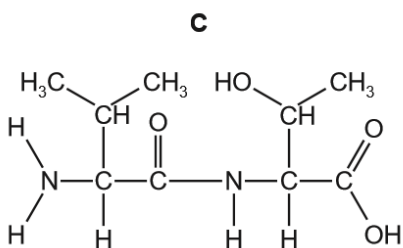
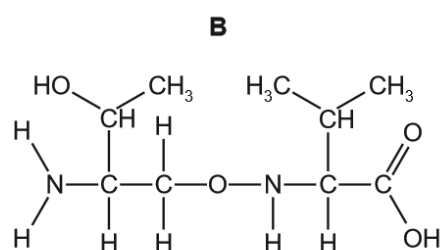
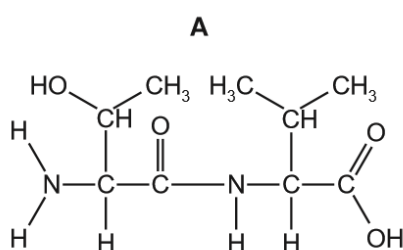


valine



threonine

Which molecular structure is threonylvaline? **ANSWER: A**



- 5 Which roles of the cell surface membrane are a result of the properties of the phospholipids?

- 1 to allow cytokinesis to occur in mitotic cell division
- 2 to allow entry and exit of oxygen and carbon dioxide
- 3 to allow the phagocytosis of a bacterium into a cell

A 1, 2 and 3

B 1 and 2 only

C 1 and 3 only

D 2 and 3 only

6 Which set of statements correctly describes haemoglobin?

	1	2	3	4
A	four polypeptide chains, each containing a haem group	iron ions can associate with oxygen forming oxyhaemoglobin	in each chain, hydrophobic R groups of amino acids point towards the centre of the molecule	at 50 % saturation, two oxygen molecules are transported by the molecule
B	polypeptide chains interact to produce a globular chain	each chain contains a haem group of amino acids surrounding an iron ion	consists of two identical alpha chains and two identical beta chains	each chain can transport an oxygen molecule
C	polypeptide chains interact to produce an almost spherical molecule	an iron ion is present within each haem group	quaternary structure has two alpha chains and two beta chains	each molecule can transport a total of four oxygen atoms
D	polypeptide chains produce a loose helical shape, which folds to form a spherical molecule	iron ions in the molecule can bind reversibly with oxygen	in each chain, hydrophobic R groups of amino acids surround the iron ion	each molecule can transport a total of eight oxygen atoms

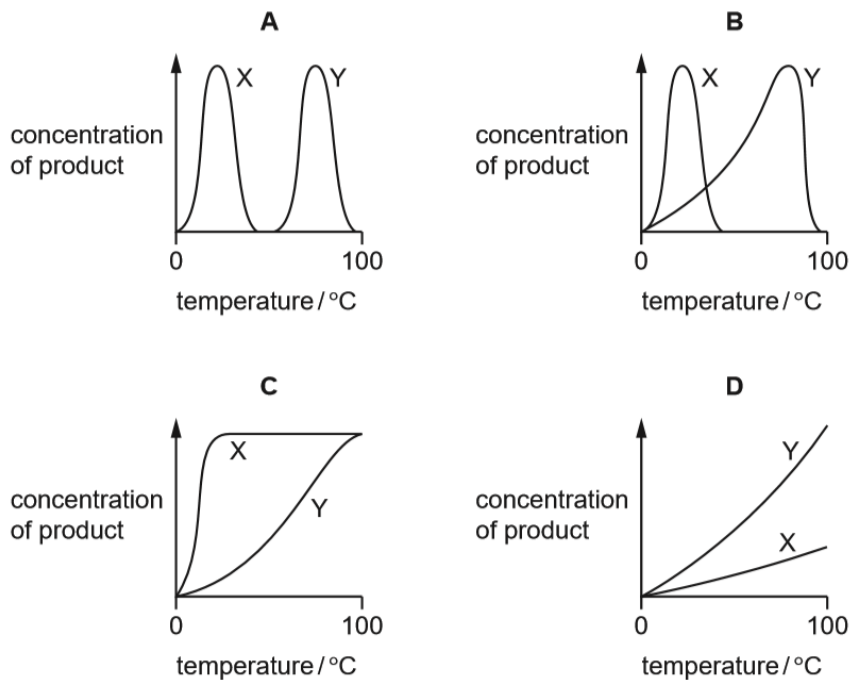
- 7 Two enzymes, X and Y, were used in an experiment.

Enzyme X was from bacteria that live in rivers and lakes at temperatures from 5°C to 20°C.

Enzyme Y was from bacteria that live in hot water springs at temperatures from 40°C to 85°C.

The experiment measured the concentration of product produced by each enzyme at temperatures between 0°C and 100°C after 5 minutes.

Which graph shows the results? **ANSWER: B**



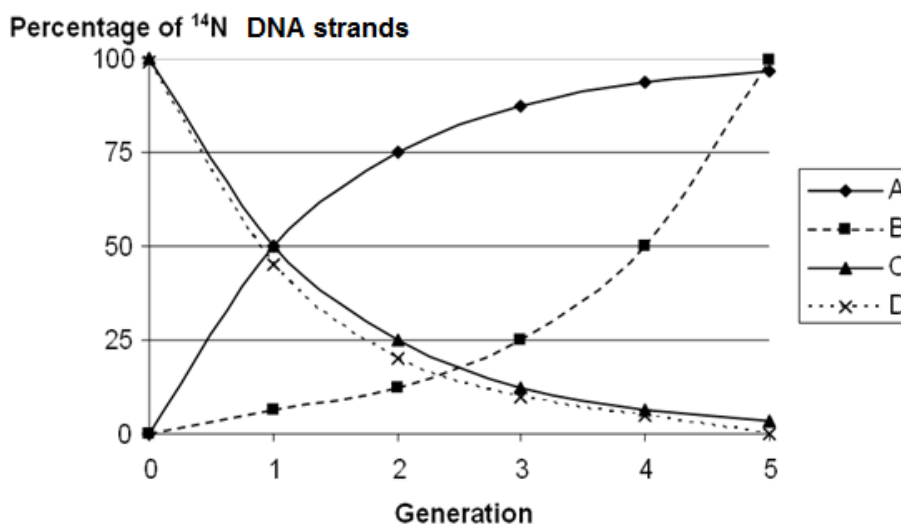
- 8 Within its own environment a particular cell line cannot be induced to produce a cell from a different cell line.

Which statement explains this?

- A Genes not required for a particular cell line are methylated.**
- B Genes not required for a particular cell line are removed by enzymes.
- C Only pre-mRNA that is required for a particular cell line is processed.
- D Stem cells have only the genes required for their particular cell line.

- 9 Bacteria were cultured in a medium containing heavy nitrogen (^{15}N) until all DNA was labelled. These bacteria were then grown in a medium containing only normal nitrogen (^{14}N) for 5 generations. The percentage of ^{14}N DNA strands in each generation was estimated.

Which curve provides evidence that DNA replication is semi-conservative? **ANSWER: A**



- 10 An unidentified single-stranded molecule was described as having the following features.
- complementary base pairing along some of its length
 - an area that can attach to a ribosome
 - a site to which a specific amino acid attaches

What is the unidentified molecule?

- A ribosomal RNA
- B messenger RNA
- C RNA polymerase
- D transfer RNA**

11 Some antibacterial drugs can affect the synthesis of proteins.

antimicrobial drug	rifampicin	streptomycin	tetracycline
mode of action	binds to RNA polymerase	genetic code misread during translation	prevents binding of tRNA to ribosome

Which is the correct set of immediate effects of these drugs?

antimicrobial drug	rifampicin	streptomycin	tetracycline
A	defective protein synthesised	mRNA does not bind to ribosome	amino acids not added to growing chain
B	mRNA not synthesised	defective protein synthesised	amino acids not added to growing chain
C	mRNA not synthesised	mRNA does not bind to ribosome	transcription prevented
D	transcription prevented	defective protein synthesised	mRNA does not bind to ribosome

12 Which statement about prokaryotes and chloroplasts is correct?

- A** Prokaryotes and chloroplasts have circular DNA where genes carrying the code for cell walls are located.
- B** Prokaryotes and chloroplasts have 70S ribosomes that are the sites for translation and polypeptide synthesis.
- C** Prokaryotes and chloroplast have an outer membrane and a separate inner, folded membrane where ATP synthesis occurs.
- D** Prokaryotes and chloroplast have double-stranded linear DNA where genes carrying coded information are located.

- 13** Human immunodeficiency virus (HIV) is a retrovirus. After infecting a host cell, viral DNA is produced which is incorporated into the DNA of the host cell. The modified host genome now codes for the production of new HIV particles.

Which could be used as a potential treatment to slow down the spread of HIV?

- 1 inhibitors of restriction endonucleases
- 2 inhibitors of reverse transcriptase
- 3 restriction endonucleases
- 4 reverse transcriptase

A 1 and 4 only

B 1 only

C 2 and 3 only

D 2 only

- 14** Which of the following does not occur during bacterial conjugation?

A direct contact between donor and recipient cells

B shortening of the pilus

C unidirectional transfer of both DNA strands

D enzymatic cleavage of one strand at the origin of transfer

15 Transcriptional control in eukaryotic cells can be accomplished at several levels.

What may be involved in such control?

- 1 The same combination of DNA binding proteins regulate the activity of all genes.
- 2 Enhancers may be involved in the promotion as well as regulation of gene transcription.
- 3 Phosphorylation of transcriptional factors by a kinase may occur.
- 4 Enhancers may be some distance from the promoter sites they control.

- A** 1, 2, 3 and 4
- B** 1, 2 and 3 only
- C** 1, 3 and 4 only
- D** 2, 3 and 4 only

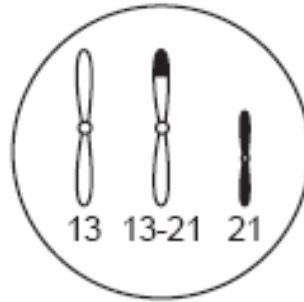
16 Multiple copies of a wanted DNA fragment can be made by the polymerase chain reaction (PCR).

Which description of this procedure is not correct?

- A** After 'n' turns of the PCR cycle, up to 2^n copies of the wanted DNA are produced.
- B** Using a heat-stable enzyme, such as Taq polymerase, means that the enzyme does not lose activity over time.
- C** Using an enzyme with a high optimum temperature allows DNA polymerisation above the annealing temperature.
- D** Using specific primers means that only the wanted DNA is replicated.

- 17** Down's syndrome can be caused by a trisomy of chromosome 21, but can also result from the translocation of chromosome 21 into chromosome 13, forming a single chromosome 13-21.

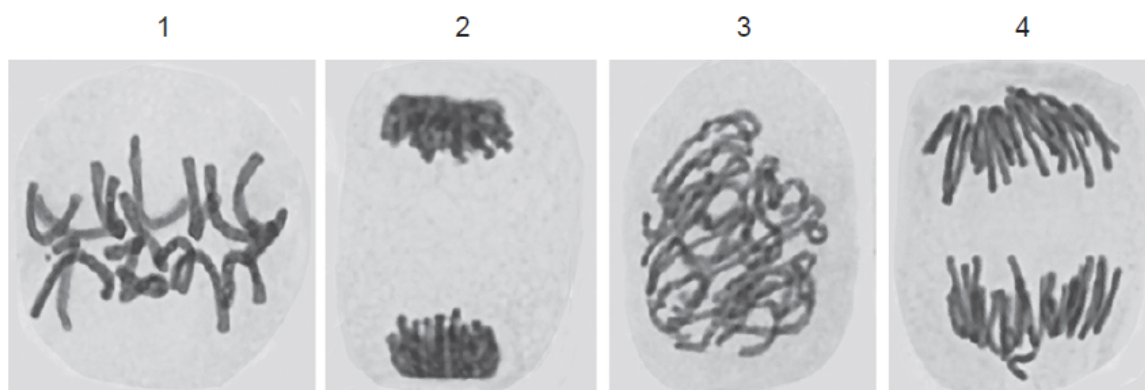
The diagram shows chromosomes 13 and 21 in the nucleus of a diploid ($2n$) testis cell from a phenotypically normal male carrier of a 13-21 translocation. This cell has a chromosome number of 45.



Which is **not** a likely outcome of fertilisation of normal oocytes by sperm from this male?

	chromosomes in sperm	embryo
A	13 and 21	$2n = 46$ normal phenotype
B	13-21	$2n = 45$ normal phenotype
C	13-21 and 21	$2n = 46$ Down's syndrome
D	13-21 and 21	$2n = 47$ Down's syndrome

18 The photomicrographs show cells in various stages of the cell cycle.



Which cells contain twice as many DNA molecules as a cell from the same organism after cytokinesis?

- A** 1, 2, 3 and 4
B 1, 2 and 4 only
C 1 and 3 only
D 2 and 4 only

19 Gene mutations in either the *BRCA1* or the *BRCA2* genes are responsible for the majority of hereditary breast cancer in humans.

The proteins produced by the two genes migrate to the nucleus where they interact with other proteins, such as those produced by the tumour suppressor gene, *p53*, and the DNA repair gene, *RAD51*.

Which combination of gene activity is most likely to result in breast cancer? **ANSWER: D**

	gene		
	<i>BRCA1</i> or <i>BRCA2</i>	<i>p53</i>	<i>RAD51</i>
A	✓	✓	✓
B	✓	✓	x
C	✓	x	✓
D	x	x	x

key

✓ = gene produces normal protein

x = gene produces abnormal protein
or no protein

- 20 What maximum number of different genotypes and phenotypes are possible among the children of a mother with blood group A and a father with blood group B?

	genotype	phenotype
A	2	2
B	2	4
C	4	4
D	4	2

- 21 A test cross resulted in these recombinants:

$$\frac{tB}{tb} \quad \frac{Tb}{tb}$$

Which was the parental test cross?

- A $\frac{TB}{tb} \times \frac{tb}{tb}$
- B $\frac{TB}{tB} \times \frac{tb}{Tb}$
- C $\frac{Tb}{tB} \times \frac{tb}{tb}$
- D $\frac{TB}{tb} \times \frac{TB}{tb}$

- 22 Isolated chloroplasts, suspended in buffer solution, are often used to study the light dependent stage of photosynthesis.

During this stage, electrons (e^-) are transferred by carriers and provide energy so that a proton (H^+) gradient can be formed. Protons diffuse through membrane proteins that are linked to synthase enzymes.

Three compounds that can be added to isolated chloroplasts are:

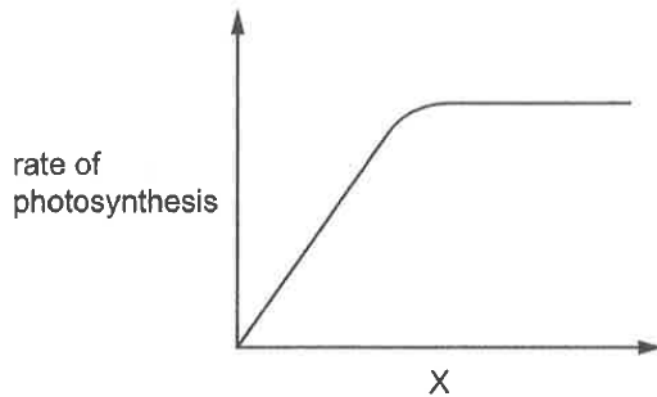
- 1 DCMU, which inactivates a carrier that accepts electrons from photosystem II
- 2 DCPIP, which can act as a final electron acceptor
- 3 ammonium hydroxide solution, which absorbs protons

Which compounds, when added separately to isolated chloroplasts, would allow the light dependent stage of photosynthesis to occur and which would inhibit it?

	allow	inhibit
A	1	2 and 3
B	1 and 3	2
C	2	1 and 3
D	2 and 3	1

- 23 The rate of photosynthesis in pondweed was measured when one variable was changed and all others were standardised.

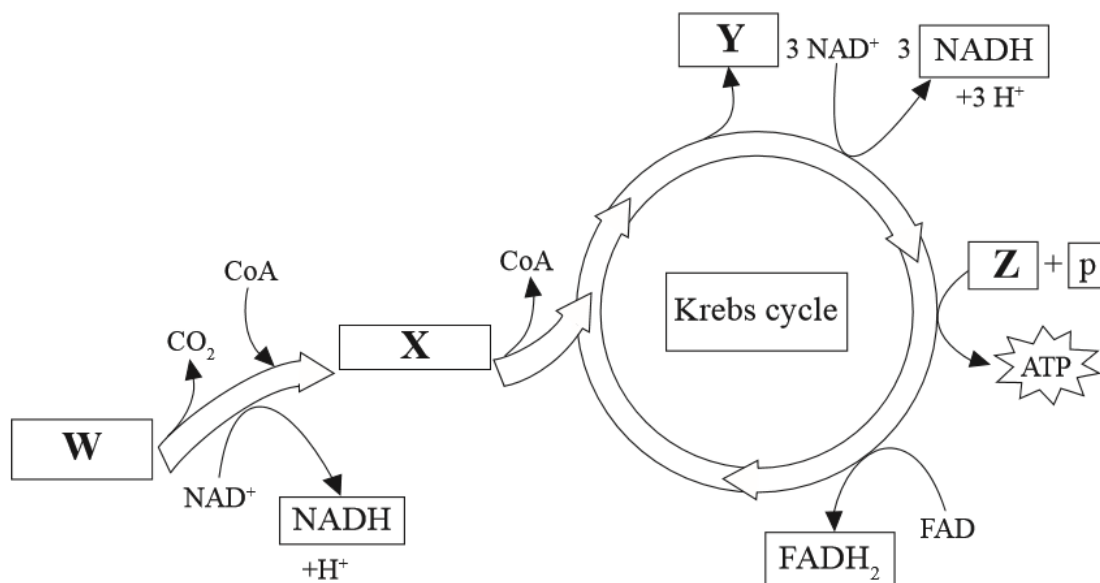
The graph shows the rate of photosynthesis at different values of a variable, X.



Which variables could be represented by X?

- 1 carbon dioxide availability
 - 2 light intensity
 - 3 oxygen availability
 - 4 temperature
 - 5 leaf area exposed to direct light
- A 1, 2 and 5
- B 1 and 2 only**
- C 2, 4 and 5
- D 3 and 4

- 24 The diagram below shows the link reaction and stages of the Krebs cycle. Which molecules are represented by the letters W, X, Y and Z?



	W	X	Y	Z
A	acetyl CoA	carbon dioxide	ADP	pyruvate
B	pyruvate	acetyl CoA	carbon dioxide	ADP
C	ADP	carbon dioxide	acetyl CoA	pyruvate
D	acetyl CoA	pyruvate	carbon dioxide	ADP

- 25 The function of phosphatases in signal transduction is to

- A move the phosphate group of the transduction pathway to the next molecule of a series.
- B prevent a protein kinase from being reused when there is another extracellular signal.
- C amplify the second messengers such as cAMP. / amplify the transduction signal so it affects multiple transducers.
- D inactivate protein kinases and turn off the signal transduction.

26 Which statements are acceptable parts of Darwinian evolutionary theory?

- 1 Advantageous behaviour acquired during the lifetime of an individual is likely to be inherited.
- 2 In competition for survival, the more aggressive animals are more likely to survive.
- 3 Species perfectly adapted to a stable environment will continue to evolve.
- 4 Variation between individuals of a species is essential for evolutionary change.

A 1, 2 and 4 only

B 2 and 3 only

C 3 and 4 only

D 4 only

27 Before the settlement of California in the 1800s, the elk population was very large. By about 1900 there were only a few dozen elk left.

Owing to protection, there are now about 3000 elk living in a small number of isolated herds.

Unfortunately, some of the elk in all the herds have difficulty grazing due to a shortened lower jaw.

Which statements best explain this?

- 1 The early settlers only hunted elk that could graze.
- 2 There was a mutation affecting jaw size in one of the herds.
- 3 There is random mating within each herd.
- 4 The current elk population demonstrates a founder effect.
- 5 There was directional selection favouring short jaws.

A 1, 2 and 4 only

B 2, 3 and 5 only

C 2 and 5 only

D 3 and 4 only

- 28 Darwin's view of the process of evolution to form new species (speciation) has been reinforced by more recent discoveries in genetics and cell biology.

In this view, which sequence of events is considered most likely to lead to speciation?

A	adaptation to population → competition and predation leading to natural selection → behavioural isolation → sympatric speciation
B	adaptation to population → competition and predation leading to natural selection → behavioural isolation → allopatric speciation
C	competition and predation leading to natural selection → geographical isolation → adaptation of isolated populations → sympatric speciation
D	competition and predation leading to natural selection → geographical isolation → adaptation of isolated populations → allopatric speciation

- 29 Apart from the ABO blood groups, humans can also be Rhesus positive or Rhesus negative.

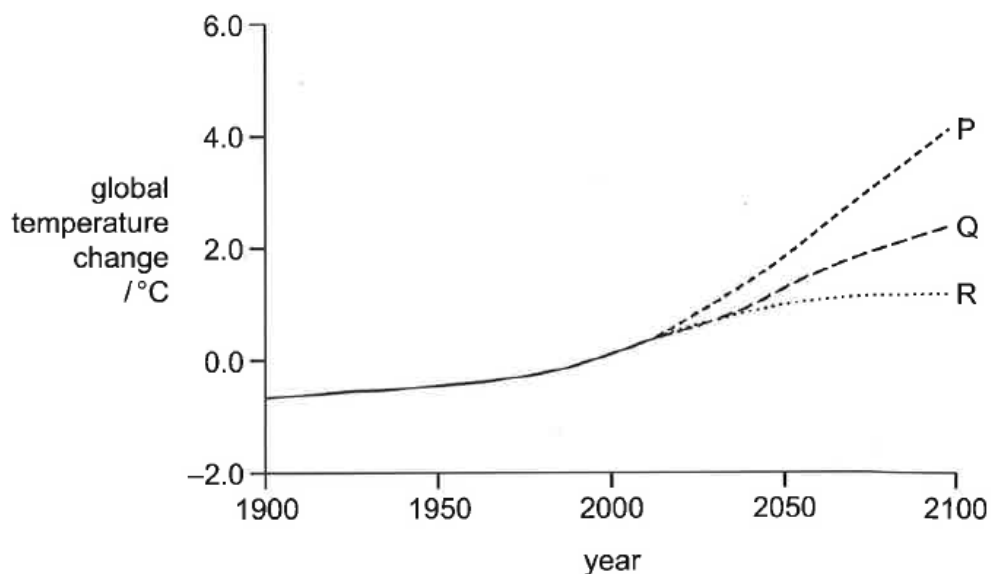
People with the Rhesus antigen are Rhesus positive. When a Rhesus negative person is given Rhesus positive blood in a transfusion there is no problem. However, a second transfusion of Rhesus positive blood to this Rhesus negative person will result in a reaction between the two types of blood.

Which statements explain this?

- 1 A Rhesus negative person naturally has anti-Rhesus antibodies.
- 2 Exposure to Rhesus antigen causes anti-Rhesus antibody production.
- 3 B-cells make anti-Rhesus antibodies.
- 4 Anti-Rhesus antibody production begins after the second exposure.

- A** 1, 2, 3 and 4
B 1 and 2 only
C 2 and 3 only
D 3 and 4 only

- 30 The graph shows the predicted change in global temperatures using three different models, P, Q and R. Model Q assumes that no new factors act to influence the rate of climate change.



The predictions based on models P and R can be explained using some of the following statements.

- 1 An increased global temperature and reduced rainfall will lead to an increase in forest fires.
- 2 Permanently frozen soil and sediment in the Arctic will begin to thaw as global temperatures rise.
- 3 Rising sea temperatures will cause increased growth of photosynthetic algae.
- 4 Rising sea temperatures will reduce the solubility of greenhouse gases in the oceans.

Which of these statements support predictions P and R?

	statements that support prediction P	statements that support prediction R
A	1, 2 and 4	3
B	1 and 3	2 and 4
C	2	1, 3 and 4
D	3 and 4	1 and 2

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